

When Truth Loses Its Source: Provenance-Aware Memory for Socially Distributed Agents

Anonymous submission

Abstract

LLM agents increasingly coordinate by talking to one another, and a growing body of work shows that information *spreads* through agent societies. We ask a reliability question that spread does not answer: when a ground-truthed fact changes, does the society come to *hold* the new truth, or merely *say* it? With a social-fidelity probe that separately measures what agents hear, say, and later hold, we find a robust speech–belief dissociation: an authoritative source drives agents to utter the current value while their held belief stays at baseline. No natural lever fixes this—not model capability, connectivity, memory swaps, richer personas, or a persistent source; only direct broadcast, an overwrite control, succeeds. The mechanism is path-dependent entrenchment: the version established early and broadly wins. We then localize the repair to a single architectural choice—integrating heard claims by *provenance* (source and version) rather than by frequency. Provenance-aware memory (PROV) is the only memory architecture that lifts held truth, and it holds across scenarios, fact types, topologies, and model families ($n=8$). Hardened into an auditable architecture (APM) with origin anchoring, corroboration, and abstain, it remains the only architecture that survives an adversarial liar (zero hijack, against PROV’s collapse) and settles to a realistic, non-saturating equilibrium under sparse communication—never confidently wrong.

Introduction

A true update can become unreliable without ever becoming intentionally false. In human societies a message may be accurate when first announced, but after a few retellings people keep the content while losing track of who said it, when it was updated, and whether it superseded an older version. This is the telephone-game failure: the social channel does not merely transmit content, it also drops source, timing, and authority cues.

LLM-agent societies reproduce this failure in an instrumented setting. Unlike a human rumor chain, an agent society exposes every utterance, every injected update, every memory snapshot, and every final answer—letting us separate three objects that are usually conflated: **reach** (whether the current update appears in the social transcript), **speech**

(whether agents later utter the current value), and the **held answer** (whether agents answer with the current value when probed after transmission).

Generative-agent work establishes that information can spread through an agent society (Park et al. 2023; Li et al. 2023; Wu et al. 2023; Zhou et al. 2024). Spread, however, is not fidelity. A correction, schedule change, or policy update can reach the conversation stream yet fail to become the society’s later answer. This is the distinction between reach and fidelity, long drawn in misinformation research (Lazer et al. 2018; Vosoughi, Roy, and Aral 2018). Our central finding is that reach and speech improve *without* durable held truth—so “information spread” is not a sufficient evaluation target for agent societies.

We make three contributions:

- a **measurement shift**: a social-fidelity probe that separates what agents hear, say, and later hold after a ground-truthed update;
- a **diagnosis**: a speech–belief dissociation driven by path-dependent entrenchment, which no natural lever repairs; and
- a **cure and an architecture**: integrating by provenance rather than frequency is the lever that works, and hardened as APM it is the only memory architecture that stays interpretable, survives an adversarial liar, and yields a realistic non-saturating equilibrium.

Our working hypothesis is that truth decays not merely because content changes, but because content loses its source. Stripped of source and version, “Saturday at the porch” and “Sunday at the community center” compete as undifferentiated text, and the old value can win simply because it was established early, repeated often, or retrieved more easily. Provenance changes the social state representation: an agent can ask not only “what have I heard?” but “who said it, is it the latest authoritative version, and does it supersede the old one?”

Related Work

Agent societies and communicative agents. Generative Agents (Park et al. 2023), CAMEL (Li et al. 2023), AutoGen (Wu et al. 2023), and SOTOPIA (Zhou et al. 2024) establish the setting: agents talk, remember, coordinate, and

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

simulate social environments. We pose a missing reliability question within it—the fidelity of a *changing* held fact.

Misinformation, rumor, reach vs fidelity. Rumor and continued-influence research separates spread from truth and shows that corrections often fail to displace stale beliefs (Lewandowsky et al. 2012; Thorson 2016). LLM-agent rumor simulations are close neighbors, but they typically measure propagation or vulnerability; we measure held-fact fidelity after a truth change. The closest concurrent neighbor (Xie et al. 2026) also studies provenance-style cures and adoption-vs-repetition; we lead instead on the explicit, separately elicited say/hold dissociation, a head-to-head decentralized-memory comparison, and adversarial robustness.

Factuality and measurement. Treating truthfulness as a semantic behavioral target rather than surface fluency is now standard (Lin, Hilton, and Evans 2022; Manakul, Liusie, and Gales 2023; Farquhar et al. 2024). Our three-way verdict mirrors fact verification (Thorne et al. 2018) and the efficacy-vs-locality structure of knowledge editing (Meng et al. 2022, 2023).

Memory, correction, recursive degradation. Memory systems (Chhikara et al. 2025; Xu et al. 2025; Zhong et al. 2024) explain why exposure is not retention; belief revision and truth maintenance (Alchourrón, Gärdenfors, and Makinson 1985; Doyle 1979) and source-credibility opinion dynamics (DeGroot 1974) motivate provenance; and model-collapse work (Shumailov et al. 2023) offers an analogy for recursive degradation—though ours is communication-time degradation, not training-time collapse.

Method: The Social-Fidelity Probe

Setup. A society of $N=25$ LLM agents on a contact graph organizes an event. At round 1 the event’s time and place *change* from a stale value (A) to a current value (B), and the update is injected into a single source agent. Agents then meet pairwise over rounds (a `meetings` parameter sets how many encounters each agent has per round, i.e. connectivity), converse, store what they hear, and retrieve memory to condition later utterances. We log every utterance (SAID) and, at the end, privately interview every agent about the current fact (HELD): Inject $\rightarrow$ Propagate $\rightarrow$ Measure.

The current / stale / unknown verdict. Because the fact’s truth value changed, we classify each held answer three ways, mirroring recognized evaluation structures: **current** (holds the new value—edit *efficacy* / SUPPORTED), **stale** (holds the superseded value—a *locality* failure / continued-influence), and **unknown** (holds neither—a retention failure / NOT-ENOUGH-INFO). The three-way split is load-bearing: separating *stale* (active corruption) from *unknown* (information loss) is essential, because interventions act on them differently—scaling capability, for instance, converts loss into confident corruption. A semantic LLM judge reproduces the keyword verdict (99–100% agreement), so the categories are not a keyword artifact.

HEARD / SAID / HELD. The probe’s leverage comes from separating what reaches an agent, what it utters, and what it later holds. “Held belief” is operational—the answer

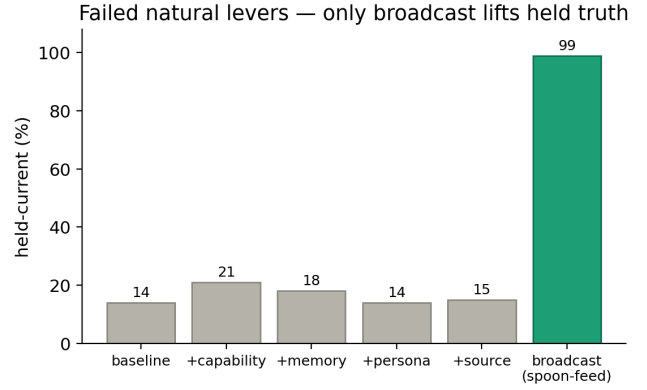


Figure 1: Failed natural levers. Capability, memory, persona, and an authoritative source all leave held truth at baseline; only broadcast (an overwrite control) succeeds.

Table 1: Failed-lever evidence ledger (held-current; ns = not significant).

Lever	Contrast	Held-current
Capability	mini→gpt-5.5	16→21% (ns)
Connectivity	meetings 1/2/3	ratio ~1.0 (ns)
Memory	GA vs. smga3g	$\Delta \approx 0$ (ns)
Persona	thin vs. thick	12 vs. 14% (ns)
Source	vs. baseline	$\Delta +0.8$ (ns)
Broadcast	vs. baseline	$\Delta +21.8 \rightarrow 99\%$

an agent gives when probed; we make no claim about literal mental states.

Results

Social Fidelity Decays, and No Natural Lever Repairs It

In the core task the current value reaches the conversation stream but does not reliably become the society’s held answer. Over a long horizon the dynamics are sharper still: current truth can rise early, peak near round 6, then decay toward a low steady state by round 30—so truth can appear transiently and still lose, and the round-5 snapshot slightly overstates retention.

Figure 1 and Table 1 rule out every intuitive repair. Capability (16→21% across mini→gpt-5.4→gpt-5.5), connectivity (roughly neutral in the powered rerun—the early “connectivity amplifies corruption” cell was an underpowered outlier, now retracted), a memory swap (the apparent authoritative source all leave held truth at the ~14–21% baseline. Only broadcasting the current value into **every** agent succeeds (99%)—an overwrite-style positive control that bypasses social transmission, not a social cure. Agents can clearly output the truth; the problem is that ordinary social communication does not reliably install it as held belief.

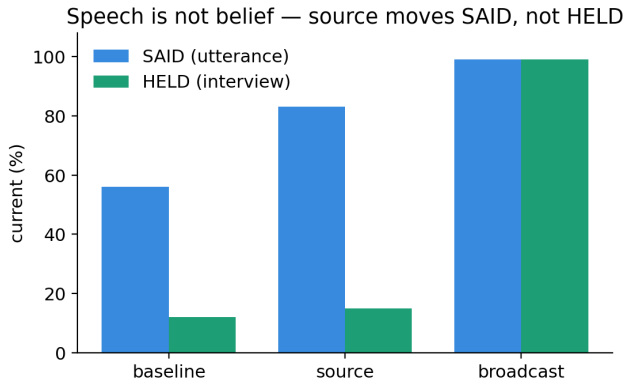


Figure 2: Speech is not belief. A source pushes SAID to 83% while HELD stays at 15%; even baseline says more than it holds; only broadcast aligns the two.

Speech Is Not Belief

Figure 2 is the centerpiece. Separating SAID (final-round utterances) from HELD (private interviews) exposes a dissociation: under a persistent authoritative source, agents reach 83% current in what they **say** while their **held** belief stays near baseline (15%). The gap is not confined to the source condition—even baseline agents say the current value more than they hold it (SAID 56% / HELD 12%)—and only broadcast aligns the two (99/99). Reach is not belief: a fact can be visible in the transcript while failing as held state. The pattern replicates across three scenarios and survives thick personae.

The Mechanism Is Entrenchment, Not Recency

Two stories could explain a sticky stale belief. If *recency* drove belief, a late all-agent broadcast just before the probe should work; if *repetition* drove it, a single early all-agent broadcast should fail. The opposite happens (Figure 3): an early (round-1) all-agent correction succeeds (100%), while an identical late (round-5) one fails (0%). What governs the outcome is whether the truth becomes the early, broad version that later conversation reinforces. The stale value starts with incumbent advantage, so a narrow or late correction can be uttered without ever taking over the society’s held state. We call this path-dependent entrenchment.

The Cure: Integrate by Provenance, Not Frequency

The diagnosis points to a precise repair. Decay arises because integration is by frequency, so the stale incumbent wins; we integrate by *provenance* instead. Each claim carries a version (its origin round) as conversation metadata; an agent holds the highest version it has *heard* and re-broadcasts that versioned belief, so the latest authoritative version supersedes the merely frequent older one. This is *fair*, not broadcast in disguise: PROV never overwrites an agent—the current value must still arrive through conversation, and an agent that never hears it stays stale or unknown. PROV changes only how an agent integrates and relays what it locally hears. It is also principled: it directly negates the diagnosed mechanism, it is normatively correct for a changing fact (frequency is the

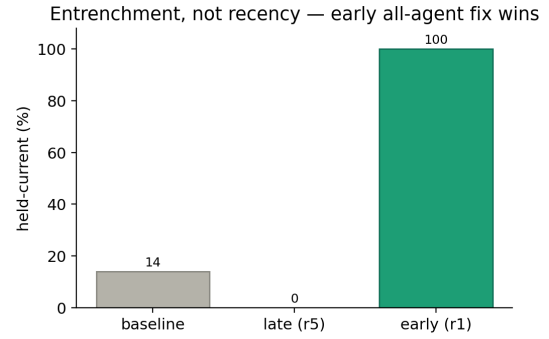


Figure 3: Entrenchment, not recency. An early all-agent correction succeeds (100%) while an identical late one fails (0%).

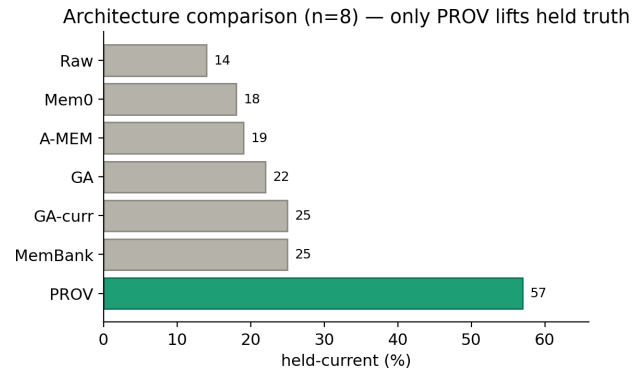


Figure 4: Architecture comparison ($n=8$). Recognized memories all fail (14–25%); only PROV lifts held truth (57%).

wrong cue—an old value repeated often is still old), and it instantiates classic ideas—truth maintenance and belief revision (Doyle 1979; Alchourrón, Gärdenfors, and Makinson 1985) and source-weighted opinion dynamics (DeGroot 1974)—in the social channel.

Figure 4 places PROV beside recognized memory architectures under one harness. Raw (14), Mem0 (18), A-MEM (19), GA (22), GA-curr (25), and MemBank (25) all fail; only PROV lifts held truth (57%). The decisive lever is the listener-side **integration rule**, not a larger or fancier memory.

The cure is not a single-condition artifact (Table 2): PROV wins across three scenarios, a numeric fact change (dues \$40→\$60), and contact topologies, with the margin *widening* on harder structured networks.

The Cure Survives Capability

Figure 5 repeats the comparison on a different model family (DeepSeek-V4-Flash) at full power ($n=8$). GA is statistically flat-and-failing across models (mini 21.5%, DeepSeek 15.5%, overlapping CIs), while PROV wins on both (57% / 64%) with CIs disjoint from GA. The cure survives capability—the mirror image of “the phenomenon survives capability.”

Table 2: Generality and capability (held-current %, GA vs. PROV).

Axis	Condition	GA	PROV
Model	mini	21.5	57
	DeepSeek	15.5	64
Scenario	repair_drive	22	57
	book_club	47	69
	carpool	18	59
Fact type	dues \$40→\$60	28	59
Topology	random	22	93
	ring	6	68
	smallworld	10	83

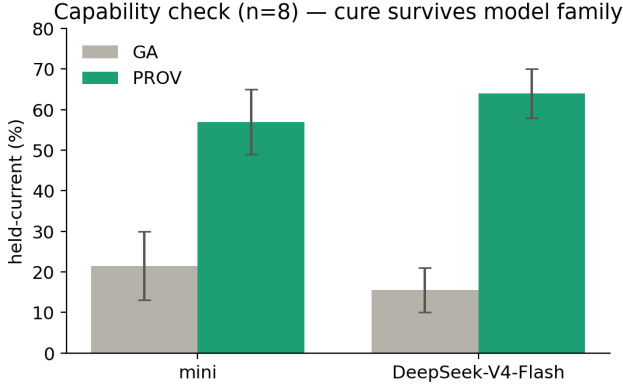


Figure 5: Capability check ($n=8$). GA is flat-and-failing across model families; PROV wins on both with disjoint CIs. Error bars are 95% CIs.

From Lever to Architecture: APM

Naive PROV identifies the right lever but is not yet a deployable architecture: it adopts the highest version blindly—a liar can claim version 999 and hijack the society—and it carries only a version, not an auditable chain. We harden it into **APM** (Auditable Provenance Memory) with three guards, each closing one gap. *Origin anchoring*: an `auth` flag is minted only at the source and relayed faithfully, so unauthenticated forgeries are rejected. *Chain corroboration*: a value is committed only after K independent sources support it, counting paths rather than value-frequency—the fix for the failure mode in which systematic garble corroborates the stale value. *Abstain*: the agent holds *unknown* until that bar is met. Because every committed belief carries a complete source→version→path trace, APM is auditable by construction—something black-box memories cannot offer.

Without an adversary, $\text{APM} \approx \text{PROV}$: interpretability is nearly free ($\sim 64\%$ on DeepSeek, with 60–76% of agents holding a fully traceable belief). Its reason to exist appears under attack (Figure 6). A liar broadcasting a forged high-version stale claim collapses PROV from 57% to 33% and hijacks 43 of 75 agents (stale 57%); APM is unmoved (64%) with **zero hijack** (stale 0), its origin anchoring rejecting the forgery society-wide. Equivalent without an adversary, APM

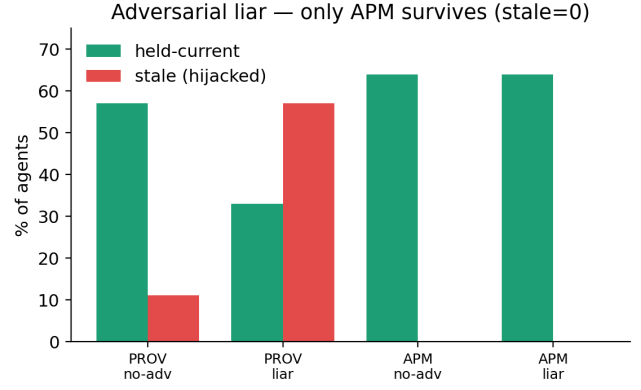


Figure 6: Adversarial liar. PROV collapses (held 33%, stale 57%) while APM is unmoved (held 64%, stale 0).

Table 3: Architecture comparison including APM. Held-current is mini/ $n=8$ for recognized memories; APM no-adversary is on DeepSeek; the under-liar column is mini.

Memory	Held	Auditable	Under liar
Raw	14	no	—
Mem0	18	no	—
A-MEM	19	no	—
GA	22	no	hijacked
MemBank	25	no	—
PROV	57	half	33 (stale 57%)
APM	≈64	yes	64 (stale 0)

is the only architecture left standing with one (Table 3).

APM Does Not Saturate

One might worry that any sticky provenance memory simply floods to 100%. That ceiling is a communication-model artifact (every-utterance, lossless broadcast), not a property of provenance integration. Figure 7 sweeps communication density—how often agents actually mention the fact. APM traces a smooth, fully monotonic S-curve: held-current climbs from 11% (mention 0, where only the source knows) to 87% (mention 0.25–0.3), and reaches 100% only at the unrealistic dense limit (0.5+). The 40% value at the sparse end is an honest floor, not a ceiling. Crucially, stale stays 0 at every density: under hard sparse communication no method spreads truth widely (the breadth advantage over GA narrows), but APM still guarantees that the informed are never misled, which GA does not. Its safety is independent of how far truth spreads.

Discussion and Limitations

What we claim. Social truth maintenance requires preserving provenance. PROV demonstrates the value of provenance-aware memory; the failure of every other lever shows the repair lives at a specific locus—listener-side integration—not in capacity. APM shows that provenance

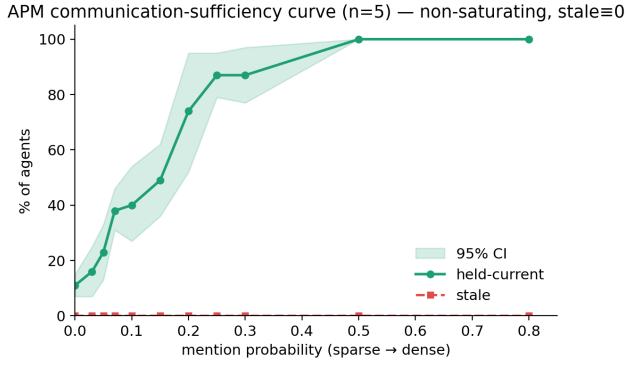


Figure 7: APM communication-sufficiency curve ($n=5$). Held-current rises monotonically with communication density and saturates only at the unrealistic dense limit; stale stays $\equiv 0$ throughout. Shaded band is the 95% CI.

can be hardened into an interpretable, adversary-resistant, non-saturating architecture.

Is structured PROV cheating? This is the central reviewer risk, and the answer is no on both sides. PROV is *not* broadcast: it still propagates through meetings, helps only agents the update reaches, and is weakened by loss, garble, sparse mention, horizon, and topology. But it is *not* fully naturalistic either—it assumes source and version survive the relay as structured metadata, so it is best read as an idealized protocol or upper bound. A companion result shows that ordinary LLM dialogue does *not* spontaneously preserve source/version, whereas an explicit attribution norm can repair held truth—a text-only upper bound that is visibly protocolized. The engineering lesson is that agent societies may need not larger memories but social-memory interfaces that preserve provenance through communication.

We did not invent provenance. Provenance is a classic idea from truth-maintenance and belief revision. Our contribution is to (i) show that default decentralized agent memories systematically lose it, (ii) localize the repair to the integration rule, (iii) quantify it through the say/hold dissociation, and (iv) harden it into an auditable, adversary-tested architecture.

Limitations. All experiments are simulations, and “held belief” is operational. APM’s headline cells are $n=3-5$ pilots (versus $n=8$ for the diagnosis and the capability check); under sparse communication its breadth advantage over GA is suggestive (overlapping CIs) while its safety advantage (stale 0 vs. 30) is decisive. APM’s corroboration knob K trades flooding ($K=1$, PROV-like) against over-conservatism ($K \geq 2$ deadlocks without relay-before-commit), and value fidelity remains a dependency—heavy garble still degrades it. We deliberately disable individual forgetting so that the failure is attributable to the integration rule rather than to memory loss; whether forgetting could itself thaw entrenchment is a question we leave to a sequel.

Human-society reading. The telephone effect is not always a defect—for descriptive social simulation, selective absorption and provenance loss are features to be reproduced,

not bugs. Our normative claim is narrower: once an agent society must maintain a changing fact for reliable coordination, human-like telephone effects turn from feature into engineering risk.

Conclusion

In a society of LLM agents, a ground-truthed update can be *said* without being *held*: authority moves speech but not belief, and the version established early and broadly entrenches. No natural lever repairs this. The repair is a specific architectural locus—integrate heard claims by provenance, not frequency—and it holds across scenarios, fact types, topologies, and model families. Hardened into APM, provenance integration becomes interpretable by construction, the only architecture that survives an adversarial liar, and a realistic non-saturating equilibrium that is never confidently wrong. Reach is not belief; for agent societies that must track changing facts, faithfully carrying source and version is the difference between saying the truth and holding it.

References

- Alchourrón, C. E.; Gärdenfors, P.; and Makinson, D. 1985. On the Logic of Theory Change: Partial Meet Contraction and Revision Functions. *The Journal of Symbolic Logic*, 50(2): 510–530.
- Chhikara, P.; Khant, D.; Aryan, S.; Singh, T.; and Yadav, D. 2025. Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory. *arXiv preprint arXiv:2504.19413*.
- DeGroot, M. H. 1974. Reaching a Consensus. *Journal of the American Statistical Association*, 69(345): 118–121.
- Doyle, J. 1979. A Truth Maintenance System. *Artificial Intelligence*, 12(3): 231–272.
- Farquhar, S.; Kossen, J.; Kuhn, L.; and Gal, Y. 2024. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630: 625–630.
- Lazer, D. M. J.; Baum, M. A.; Benkler, Y.; Berinsky, A. J.; Greenhill, K. M.; Menczer, F.; Metzger, M. J.; Nyhan, B.; Pennycook, G.; Rothschild, D.; Schudson, M.; Sloman, S. A.; Sunstein, C. R.; Thorson, E. A.; Watts, D. J.; and Zittrain, J. L. 2018. The science of fake news. *Science*, 359(6380): 1094–1096.
- Lewandowsky, S.; Ecker, U. K. H.; Seifert, C. M.; Schwarz, N.; and Cook, J. 2012. Misinformation and Its Correction: Continued Influence and Successful Debiasing. *Psychological Science in the Public Interest*, 13(3): 106–131.
- Li, G.; Hammoud, H. A. A. K.; Itani, H.; Khizbullin, D.; and Ghanem, B. 2023. CAMEL: Communicative Agents for “Mind” Exploration of Large Language Model Society. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Lin, S.; Hilton, J.; and Evans, O. 2022. TruthfulQA: Measuring How Models Mimic Human Falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Manakul, P.; Liusie, A.; and Gales, M. J. F. 2023. SelfCheck-GPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Meng, K.; Bau, D.; Andonian, A.; and Belinkov, Y. 2022. Locating and Editing Factual Associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Meng, K.; Sharma, A. S.; Andonian, A.; Belinkov, Y.; and Bau, D. 2023. Mass-Editing Memory in a Transformer. In *International Conference on Learning Representations (ICLR)*.

Park, J. S.; O’Brien, J. C.; Cai, C. J.; Morris, M. R.; Liang, P.; and Bernstein, M. S. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*.

Shumailov, I.; Shumaylov, Z.; Zhao, Y.; Gal, Y.; Papernot, N.; and Anderson, R. 2023. The Curse of Recursion: Training on Generated Data Makes Models Forget. *arXiv preprint arXiv:2305.17493*.

Thorne, J.; Vlachos, A.; Christodoulopoulos, C.; and Mittal, A. 2018. FEVER: a Large-scale Dataset for Fact Extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 809–819.

Thorson, E. 2016. Belief Echoes: The Persistent Effects of Corrected Misinformation. *Political Communication*, 33(3): 460–480.

Vosoughi, S.; Roy, D.; and Aral, S. 2018. The spread of true and false news online. *Science*, 359(6380): 1146–1151.

Wu, Q.; Bansal, G.; Zhang, J.; Wu, Y.; Li, B.; Zhu, E.; Jiang, L.; Zhang, X.; Zhang, S.; Liu, J.; Awadallah, A. H.; White, R. W.; Burger, D.; and Wang, C. 2023. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. *arXiv preprint arXiv:2308.08155*.

Xie, Y.; Zhu, C.; Zhang, X.; Zhu, T.; Ye, D.; Qi, M.; Chen, H.; and Zhou, W. 2026. From Spark to Fire: Modeling and Mitigating Error Cascades in LLM-Based Multi-Agent Collaboration. *arXiv preprint arXiv:2603.04474*.

Xu, W.; Liang, Z.; Mei, K.; Gao, H.; Tan, J.; and Zhang, Y. 2025. A-Mem: Agentic Memory for LLM Agents. *arXiv preprint arXiv:2502.12110*.

Zhong, W.; Guo, L.; Gao, Q.; Ye, H.; and Wang, Y. 2024. MemoryBank: Enhancing Large Language Models with Long-Term Memory. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(17): 19724–19731.

Zhou, X.; Zhu, H.; Mathur, L.; Zhang, R.; Yu, H.; Qi, Z.; Morency, L.-P.; Bisk, Y.; Fried, D.; Neubig, G.; and Sap, M. 2024. SOTOPIA: Interactive Evaluation for Social Intelligence in Language Agents. In *International Conference on Learning Representations (ICLR)*.